

FADO: A Reaction Controller for Fairness-Aware Online Learning Under Concept Drift

Anonymous Author(s)

Abstract

Online learning systems in non-stationary environments must stay both predictive and fair as the data distribution evolves, yet existing fairness-aware learners optimise a fixed regulariser and ignore concept drift, while drift-adaptive streaming methods enforce no fairness. We propose Fairness-Aware Drift-Aware Online Learning (FADO), which couples the ARANYANI fair oblique decision forest [?] with a dual-threshold ADWIN detector and a reaction controller that spikes the learning rate and sharpens the soft-routing temperature on confirmed drift, annealing both back during recovery. We target the *multi-attribute* setting, optimising the *intersectional* demographic parity (DP) over the Cartesian-product subgroups of two protected attributes rather than each axis alone — necessary to avoid the fairness-gerrymandering loophole of marginal-only regularisation. On FOLKTABLES ACS Income (CA), a natural 2015 → 2017–18 cross-year shift with sex × race grouping, aggregated over 30 seeds, FADO attains the best intersectional and marginal DP against the ARF and RFR streaming baselines and a controller-free ARANYANI ablation, beating every baseline at paired Wilcoxon Holm-corrected $p < 0.001$ — including a both-tests-significant reduction over the controller-free baseline that the single-attribute setting did not surface — at a negligible ($\leq 0.3\text{pp}$, Welch $p = 0.10$) accuracy cost, more than halving the DP of ARF and RFR. We further find that the fair learners carry the highest residual intersectional-to-marginal disparity ratio, empirically confirming the gerrymandering risk intersectional optimisation removes. Direct equalised-odds optimisation and the intersectional COMPAS testbed are the subject of an ongoing companion sweep.

Keywords

fairness-aware online learning, concept drift, intersectional fairness, oblique decision forests, drift detection

1 Introduction

The increasing deployment of machine learning (ML) systems in high-stakes decision-making domains, including healthcare [?], criminal justice [?], and hiring [?], where their predictions may directly affect individuals and society makes optimizing for performance an insufficient objective; models must also satisfy fairness requirements to ensure equitable treatment across individuals or groups of individuals [?]. In this regard, among the fairness literature, *group fairness* [?] has emerged as one of the most widely studied paradigms, aiming to ensure that predictive behaviour is balanced across populations defined by sensitive attributes such as gender, race, or age. Common formulations include demographic parity, equal opportunity, and equalized odds, each imposing statistical constraints on model outcomes across groups (sometimes mutual incompatible) [?]. Most existing fairness-aware models, however, are developed under the assumption of a stationary data distribution. In practice, real-world environments are inherently

dynamic, and the statistical properties of data may evolve over time due to policy changes, behavioural shifts, seasonal effects, economic fluctuations, or other social transformations. This phenomenon, commonly referred to as concept drift, occurs when the relationship between input variables and target outcomes changes over time, causing previously learned decision boundaries to become outdated. Different forms of concept drift [?] require different adaptation strategies and can degrade performance when models fail to adjust. Drift also threatens fairness: shifting distributions alter the statistical relationships that support fairness guarantees, increasing bias against some groups. To address this challenge, the online learning and data stream mining literature has proposed numerous adaptive methods capable of detecting and responding to concept drift in real time. These approaches often rely on drift detection mechanisms combined with incremental learning strategies to maintain predictive accuracy under non-stationary conditions. Nevertheless, the majority of existing methods focus primarily on predictive utility [?], while fairness considerations remain comparatively underexplored. As a result, fairness guarantees established during training may deteriorate over time even when predictive accuracy remains relatively stable [?]. This limitation is particularly critical in high-stakes applications, where persistent fairness violations may remain undetected for extended periods, compromising trustworthiness, reinforcing societal inequalities, and causing tangible harm to affected groups. These limitations highlight an important gap in the literature and motivate the following research questions:

- **RQ1:** To what extent does concept drift affect group fairness metrics in online learning settings?
- **RQ2:** Can adaptive drift detection and response mechanisms preserve fairness guarantees while maintaining predictive performance over time?

Based on these research questions, this work investigates fairness-aware online learning under concept drift. We introduce Fairness-Aware Drift-Aware Online Learning (FADO), a framework that wraps an existing fair oblique decision forest—ARANYANI [?]—with a dual-threshold drift detector and a reaction controller that modulates the optimizer learning rate and the soft-routing temperature in response to detected change. We emphasize that ARANYANI is not contributed by this paper: it serves as the in-processing base learner around which FADO builds its drift-detection and response machinery. Our central hypothesis is that fairness degradation under concept drift can be mitigated by coupling an in-processing fairness-aware ensemble with an explicit, fairness-aware adaptive controller, allowing the model to maintain both predictive utility and equitable treatment across demographic groups in non-stationary environments.

To evaluate this hypothesis, we run a prequential, test-then-train evaluation under *intersectional* (joint two-attribute) grouping. FOLKTABLES (ACS Income, CA) contributes a natural cross-year shift between 2015 (training) and 2017–2018 (prequential test) with

sex $\times$ race as the joint protected attribute, and a controlled synthetic concept-drift testbed on COMPAS (race $\times$ sex; Section ??) is reported in companion work. FADO is compared against three streaming baselines spanning the accuracy/fairness/adaptivity design space: ARF [?], an accuracy-oriented Adaptive Random Forest with drift detection but no fairness mechanism; RFR [?], a sharpness-aware fairness-reweighted variant without adaptive drift handling; and a controller-free ARANYANI baseline, identical to the base learner inside FADO but with the drift-reaction controller disabled, which isolates the controller's contribution. The main contributions of this work are:

- FADO, a fairness-aware online learning framework that wraps a fair oblique decision forest with a dual-threshold ADWIN drift detector and a reaction controller acting on the learning rate and the soft-routing temperature;
- A *multi-attribute* (intersectional) instantiation of the node-level fairness regulariser that optimises demographic parity over the Cartesian-product subgroups of two protected attributes, motivated by the fairness-gerrymandering result that marginal-only regularisation is provably defeatable [?];
- An empirical analysis, on a natural cross-year shift on FOLK-TABLES (with a controlled concept-drift testbed on COMPAS described in Section ?? and reported in companion work), of how concept drift affects intersectional group fairness over time, comparing FADO against the ARF, RFR, and controller-free ARANYANI baselines and showing that the reaction controller delivers a both-tests-significant intersectional-DP reduction at negligible accuracy cost, while every method's residual disparity remains concentrated in the joint subgroups that marginal metrics hide.

2 Related Work

Following the most adopted taxonomy [?], in this section, the works will be divided into three categories: (i) pre-processing methods, (ii) in-processing and (iii) post-processing methods.

2.1 Pre-processing methods

Pre-processing methods aim to mitigate bias in the data before the training stage of a model. Xiong et al. [?] propose leveraging training samples reweighting to satisfy demographic parity while minimizing distributional distortion using Wasserstein distance to the original dataset. Although effective, this method does not account for concept drift, nor proves robustness to other forms of fairness violations which limits its applicability in dynamic environments. Zeng et al. [?] propose a different approach that contrasts with the aforementioned, modifying the actual data labels by treating the removal of discrimination from the training dataset as a randomized response problem. The main contribution of this work is the provision of a theoretical bridge between pre-processing and techniques to downstream Bayes-optimal classifiers and the fact that it is model agnostic: because it only alters the training labels, users do not need to modify their current pipelines. However, this method also does not account for concept drift nor its impact, and it is limited by the binary setup, meaning it only works with binary sensitive attributes and binary outcomes.

2.2 In-processing methods

In-processing methods incorporate fairness constraints directly into the learning algorithm. Three lines are especially relevant to our work.

Chowdhury et al. [?] propose ARANYANI, an online fair-learning ensemble of soft-routed oblique decision trees in which the demographic-parity penalty is enforced at the level of individual routing nodes, inducing a parameter-isolation property that keeps the per-step fairness gradient cheap and lets the ensemble track running per-group statistics rather than storing past samples. We adopt this base learner directly; its limitations — DP-only, binary protected attributes, no drift mechanism — motivate our extension.

Gomes et al. [?] propose Adaptive Random Forest (ARF), a streaming ensemble of Hoeffding Trees with per-tree warning and drift detectors that swap in background trees on confirmed drift. ARF is the standard streaming-adaptation baseline but is purely accuracy-oriented; we treat it as a representative adaptive-but-not-fair baseline.

Jiang et al. [?] model distribution shift as a bounded weight-space perturbation and optimise for worst-case fairness under that perturbation, producing the sharpness-aware reweighting (RFR) that we use as our fair-but-not-adaptive baseline.

2.3 Post-processing methods

Post-processing approaches transform an already-trained classifier's outputs. Hardt et al. [?] introduce randomised threshold adjustments that equalise group error rates and remain the standard model-agnostic baseline; Mishler et al. [?] extend the idea to counterfactual equalised odds in intervention settings (at the cost of strong causal assumptions); Small et al. [?] replace fixed randomisation regions with smooth monotonic probability curves to recover individual fairness. None of these methods touch the training loop, so none of them adapt under drift; we mention them for completeness rather than as direct comparators.

2.4 Main Limitations of Prior Work

Notwithstanding the significant contributions of the three families surveyed above, we identify four main limitations that jointly motivate this work and the design of FADO:

- **L1 – Static fairness assumption.** Once the protected-attribute conditional or the label distribution shifts, the fairness guarantees obtained at training time provide no protection at deployment;
- **L2 – Adaptive methods are accuracy-oriented.** Streaming ensembles such as ARF [?] explicitly detect and react to drift but do so purely on the predictive-error signal, with no mechanism for tracking or correcting fairness violations;
- **L3 – Existing fair online learners do not react to drift.** ARANYANI [?] provides memory-efficient online fairness, but its regulariser is non-adaptive: when the distribution moves it continues to optimise the same penalty with the same hyperparameters, and convergence to the new optimum is slow;
- **L4 – Drift-aware fair methods rely on assumptions that do not transfer to streams.** Approaches such as [?]

treat distribution shift as a bounded weight-space perturbation, which is useful in worst-case analysis but doesn't yield an actionable online detector or controller;

FADO focuses on addressing L1–L4 directly by retaining the node-level fairness regulariser of ARANYANI as the in-processing base learner, and adding on top of it (i) an ADWIN-based dual-threshold detector on the online error stream, (ii) a reaction controller that modulates the optimiser learning rate and the soft-routing temperature on warning and confirmation events, and (iii) a rolling-window fairness monitor whose statistics also drive the fairness-aware updates.

3 Problem Statement

3.1 Online Learning Setting

We consider an online binary classification problem over a data stream $\{(x_t, y_t, a_t)\}_{t=1}^T$, where $x_t \in \mathbb{R}^d$ is the feature vector observed at time t , $y_t \in \{0, 1\}$ is the binary target label, and $a_t \in \mathcal{A}$ is a sensitive attribute taking values in a finite set with $|\mathcal{A}| = K \geq 2$. Following the formulation of previous work [? ?], at each time step the learner receives x_t , produces a prediction $\hat{y}_t = \arg \max f_t(x_t)$, then observes (y_t, a_t) and updates its parameters under a prequential, test-then-train protocol. The fairness formulation introduced below supports arbitrary K , although some datasets are binarised in our study (given that most existing benchmarks only contain binary protected attributes).

Multiple protected attributes. When more than one protected attribute is observed – e.g. $a_t = (a_t^{(1)}, a_t^{(2)})$ with $a_t^{(k)} \in \mathcal{A}^{(k)}$ – we adopt the *intersectional* (Cartesian-product) instantiation of the same single-attribute formulation: we set $\mathcal{A} = \mathcal{A}^{(1)} \times \mathcal{A}^{(2)}$ and treat the joint subgroup $a_t = (a_t^{(1)}, a_t^{(2)})$ as a single index into $\mathcal{A}$ with $K = |\mathcal{A}^{(1)}| \cdot |\mathcal{A}^{(2)}|$. This choice follows the intersectional-fairness reading of group fairness in [? ?]: marginal regularisation along each attribute axis *alone* is provably defeatable by classifiers that achieve perfect per-axis demographic parity while violating subgroup parity on the intersection (the “fairness gerrymandering” construction of [?]), whereas optimising the intersectional violation directly closes that loophole by construction. The node-level regulariser of Section ?? applies unchanged at $K = |\mathcal{A}|$; the controller of Section ?? is attribute-agnostic and does not need to be re-derived. For transparency we additionally report the *marginal* per-attribute demographic-parity violations as diagnostics in Section ??, computed from the same prediction stream but with $\mathcal{A}_t$ restricted to a single attribute's groups.

3.2 Group Fairness Metrics

We focus on *group fairness*, which assesses whether model behaviour is statistically similar across protected groups. Although [?] already introduces demographic parity as the node-level regulariser we had to generalize it to support equalised odds as well as non-binary protected attributes and, for that reason, we must introduce the group-level metrics in their refactored form. Let $\mathcal{A}_t \subseteq \mathcal{A}$ denote the set of sensitive groups observed within the current evaluation window, and let $\hat{p}_a = \mathbb{E}[\hat{y} \mid a]$ be the positive-prediction rate for group $a \in \mathcal{A}_t$. We define the unweighted mean group rate

as:

$$\bar{p} = \frac{1}{|\mathcal{A}_t|} \sum_{a' \in \mathcal{A}_t} \hat{p}_{a'},$$

i.e. the average positive-prediction rate across the observed groups (a reference value against which each per-group rate $\hat{p}_a$ is compared). The *demographic parity (DP) violation* is then the worst-case deviation of any group's positive-prediction rate from this mean:

$$DP = \max_{a \in \mathcal{A}_t} |\hat{p}_a - \bar{p}|,$$

and is bounded between 0 (perfect parity) and 1 (one group always predicted positive while another never is). DP measures bias in the prediction marginal and is sensitive to differences in base rates across groups. We adopt this mean-from-average form rather than the more common pairwise gap $\max_{a,a'} |\hat{p}_a - \hat{p}_{a'}|$ because it matches the structure of the node-level fairness regulariser introduced in Section ??, which the model actually optimises. In our experiments each protected attribute is binary, so the joint subgroup index of Section ?? has $K = |\mathcal{A}| = 4$ groups and the intersectional DP is the mean-from-average maximum over those four cells. For the binary *marginal* diagnostics ($K = 2$) this form reduces to $DP = \frac{1}{2} |\hat{p}_0 - \hat{p}_1|$, equivalent up to a factor of two to the textbook pairwise gap; for the four-cell intersectional metric the mean-from-average and pairwise forms differ in general, and we report the mean-from-average form throughout because it is the quantity the regulariser minimises.

We also define *equalised odds* (EO), the label-conditional analogue that takes the worst-case group deviation $|\hat{p}_{a|y} - \bar{p}_y|$ within each true-label class $y \in \{0, 1\}$ (with $y=1$ alone giving equal opportunity); its formal definition and label-conditional regulariser, used by the deferred EO-objective study, are given in Appendix ??.

3.3 Node-Level Fairness Regulariser

To connect these group-level fairness notions with the learning objective, we follow the in-processing strategy introduced by [?] and impose fairness constraints on the local outputs of the soft-routed oblique trees that form the base learner of FADO. Let $n_{ij}(x) \in [0, 1]$ denote the soft routing decision at node (i, j) , where i indexes the depth and j the node within that depth, i.e. the probability with which an input x is routed to the right child of that node. Each node thus plays a role analogous to a per-step prediction, and the same group-fairness language can be applied to it. For demographic parity, define:

$$\bar{n}_{ij,a} = \mathbb{E}[n_{ij}(x) \mid a], \quad \bar{n}_{ij} = \frac{1}{|\mathcal{A}|} \sum_{a' \in \mathcal{A}} \bar{n}_{ij,a'},$$

where $\bar{n}_{ij,a}$ is the group-conditional average routing decision at node (i, j) (the analogue of $\hat{p}_a$ but applied to the routing signal) and $\bar{n}_{ij}$ is its unweighted cross-group mean. The *node-level deviation* for group a is then:

$$F_{ij,a} = \bar{n}_{ij} - \bar{n}_{ij,a},$$

which is the per-node analogue of the demographic parity gap, signed so that positive values indicate that group a is being routed less often to the right child than the cross-group average. The relaxed fairness-regularised objective is:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \sum_{(i,j)} \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \phi_{\delta}(F_{ij,a}),$$

where $\mathcal{L}_{\text{pred}}$ is the predictive (cross-entropy) loss, the outer sum aggregates the fairness penalty across every routing node (i, j) in every tree, the inner sum averages over protected groups, and the hyperparameter $\lambda > 0$ controls the accuracy–fairness trade-off. Following the in-processing update of the ARANYANI base learner [?], we use a Huber-style piecewise quadratic–linear surrogate ϕ_δ , specified by its subgradient:

$$\phi'_\delta(z) = \begin{cases} z, & |z| < \delta, \\ \text{sign}(z), & |z| \geq \delta, \end{cases}$$

which, up to an additive constant, corresponds to the function:

$$\phi_\delta(z) = \begin{cases} \frac{1}{2}z^2, & |z| < \delta, \\ |z| + \frac{\delta^2}{2} - \delta, & |z| \geq \delta. \end{cases}$$

This surrogate behaves quadratically near zero, so small deviations receive a smooth, well-conditioned gradient, and is linear in the tails so large (possibly outlier driven) deviations do not dominate the update. It differs from the textbook Huber loss in the slope of the linear branch (unit slope rather than δ), which produces a more aggressive correction once $|F_{ij,a}|$ exceeds the threshold δ at the cost of a gradient discontinuity at $|F_{ij,a}| = \delta$. We retain this form unchanged for compatibility with the published ARANYANI update.

The label-conditional equalised-odds regulariser $\mathcal{L}_{\text{EO}}$ is obtained by conditioning each running statistic on the true label; it differs from the DP regulariser only by an additional sum over the label class $c \in \{0, 1\}$ and a $\frac{1}{2}$ normalisation, leaving the per-node penalty, group average, and ϕ_δ surrogate unchanged. We defer its full derivation to Appendix ??, since the study reported here optimises DP (Section ??).

4 Methodology and Experimental Design

FADO combines three components (Figure ??): (i) the soft-routed oblique-tree base learner with node-level fairness regularisation, inherited from ARANYANI [?] (Section ??); (ii) a fairness-aware online update that maintains running group statistics and computes node-level fairness gradients without storing past samples (Section ??); and (iii) a dual-threshold ADWIN detector coupled with a reaction controller on the learning rate and soft-routing temperature (Section ??). Components (i)–(ii) are inherited essentially unchanged; the novelty of FADO is component (iii), the rolling-window fairness monitor driving both metrics and regulariser, and the wiring of these pieces into a single prequential loop. Sections ??–?? then describe datasets, drift scenarios, baselines, hyperparameter selection, and the evaluation protocol. Algorithm ?? states the resulting prequential loop in full; the subsections that follow detail each component.

All experiments were run in Python 3.9 across three machines: a MacBook Pro (Apple M5, 16 GB RAM); a desktop with Intel i5-8500 CPU, NVIDIA GTX 1060 Ti GPU, and 16 GB RAM; and a desktop with Intel i9-9900K CPU, NVIDIA RTX 2080 Ti GPU, and 32 GB RAM. Per-seed runs were dispatched across the three machines for wall-clock parallelism; within any given seed, all four models were executed on the same machine to preserve the seed-pinned data ordering required by the paired statistical tests (Section ??). Source code, configuration files, and seed manifests are released for reproducibility (Section ??).

Algorithm 1 FADO prequential test-then-train loop

Require: stream $\{(x_t, y_t, a_t)\}_{t=1}^T$; base LR η_0 ; multipliers β_w, β_c ; temperatures $\tau, \tau_{\text{drift}}$; ADWIN levels δ_w, δ_c ; window W ; cooldown C ; guard ν

- 1: init forest f , optimiser at η_0 , temperature τ
- 2: init detectors $D_w(\delta_w), D_c(\delta_c)$; running fairness stats S
- 3: $\text{warn} \leftarrow \text{FALSE}$; $t_{\text{last}} \leftarrow -C$; $\text{recovering} \leftarrow \text{FALSE}$
- 4: **for** $t = 1$ **to** T **do**
- 5: $\hat{y}_t \leftarrow \arg \max f(x_t)$; $e_t \leftarrow 1[\hat{y}_t \neq y_t]$ ▷ test
- 6: update windowed DP (and marginal DP) over last W
- 7: $\text{noise} \leftarrow e_t \wedge (\text{conf}(\hat{y}_t) > \nu)$ ▷ label-noise guard
- 8: **if not** noise **then** $D_w.\text{update}(e_t)$; $D_c.\text{update}(e_t)$
- 9: **end if**
- 10: **if** D_w fired **and not** warn **and** $t - t_{\text{last}} \geq C$ **then**
- 11: $\eta \leftarrow \beta_w \eta_0$; $\text{warn} \leftarrow \text{TRUE}$ ▷ pre-warm
- 12: **end if**
- 13: **if** D_c fired **and** $t - t_{\text{last}} \geq C$ **then**
- 14: $\eta \leftarrow \beta_c \eta_0$; $\tau \leftarrow \tau_{\text{drift}}$ ▷ confirm: spike + sharpen
- 15: reset D_w, D_c ; $t_{\text{last}} \leftarrow t$; $\text{warn} \leftarrow \text{FALSE}$; $\text{recovering} \leftarrow \text{TRUE}$
- 16: **end if**
- 17: **if** recovering **and** rolling-acc $\geq$ pre-drift baseline **then**
- 18: anneal $\eta \rightarrow \eta_0$ over Δ ; $\tau \rightarrow 1$; $\text{recovering} \leftarrow \text{FALSE}$
- 19: **end if**
- 20: accumulate fairness stats S from (a_t, y_t) ▷ train
- 21: $g \leftarrow \nabla(\mathcal{L}_{\text{CE}}(f(x_t), y_t) + \lambda \mathcal{L}_{\text{fair}}(S))$
- 22: optimiser step on f with g at rate η
- 23: **end for**

4.1 Fair Oblique Decision Forest

The base model is an ensemble of K soft-routed oblique decision trees of depth h (Table ??), each with $2^h - 1$ internal routing nodes $n_j(x) = \sigma((w_j^\top x + b_j)/\tau) \in (0, 1)$ (trainable w_j, b_j ; shared temperature τ) and 2^h leaves storing class logits θ_ℓ . Path probabilities $p_\ell(x)$ follow from the routing decisions via a pre-computed leaf–subtree mask; the per-tree output is $f_k(x) = \sum_\ell p_\ell(x) \theta_\ell$, and the forest aggregates per-tree softmaxes, $f(x) = \frac{1}{K} \sum_k \text{softmax}(f_k(x))$, with $\hat{y} = \arg \max_c f_c(x)$. The temperature τ is the controller’s main lever: at $\tau=1$ routing is smooth and gradient-friendly (steady state), while $\tau \rightarrow 0$ sharpens routing for decisive recovery after a confirmed drift (Section ??).

4.2 Fairness-Aware Online Updates

Following the in-processing strategy of the ARANYANI base learner [?], FADO enforces fairness at the level of individual routing nodes rather than only at the output layer. Per-step learning combines the predictive loss with the node-level fairness penalty introduced in Section III:

$$\mathcal{L}_t = \mathcal{L}_{\text{CE}}(f(x_t), y_t) + \lambda \sum_{(k,j)} \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \phi_\delta(F_{kj,a}),$$

where $\mathcal{L}_{\text{CE}}$ is the per-step cross-entropy with logits and $\phi_\delta(\cdot)$ is the piecewise quadratic–linear surrogate of Section III. The implementation supports both the DP objective $\mathcal{L}$ above and the EO objective $\mathcal{L}_{\text{EO}}$ of Section ?? as selectable optimisation targets,

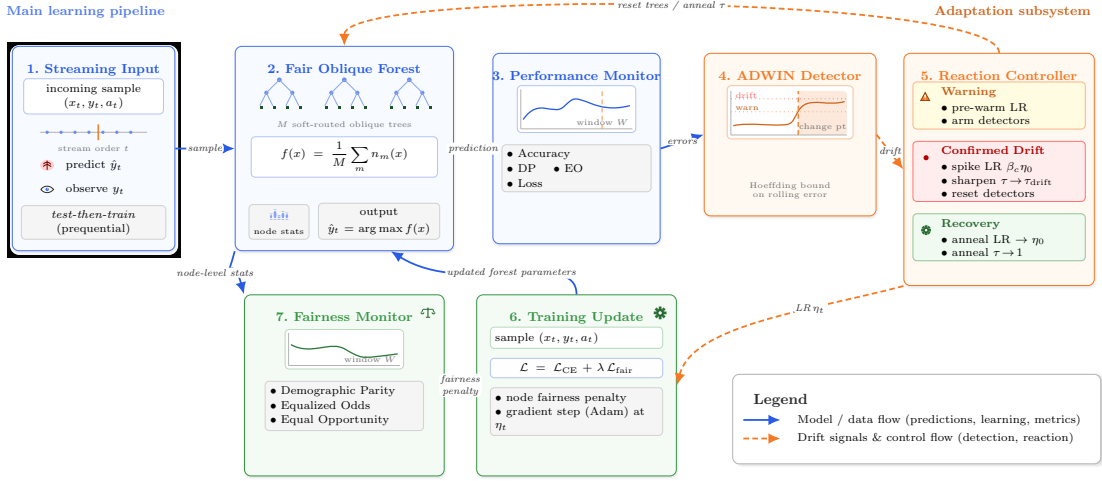


Figure 1: The FADO online learning system. The *main learning pipeline* (blue) streams each sample (x_t, y_t, a_t) through the ARANYANI fair oblique forest [?] under a prequential test-then-train protocol, monitors prequential accuracy and the intersectional fairness metrics over a rolling window of size W , and applies the fairness-regularised gradient update $\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{fair}$, which writes the updated parameters back to the forest. The *adaptation subsystem* (orange) runs the dual-threshold ADWIN detector and the reaction controller, which on a confirmed drift spikes the learning rate η_t and sharpens the soft-routing temperature τ_t , then anneals both back during recovery. Solid blue arrows denote the model/data flow; dashed orange arrows denote drift signals and control flow.

and both fairness statistics are computed and monitored online throughout the prequential stream regardless of which one is being optimised. The study reported here fixes the optimisation target to the demographic-parity regulariser $\mathcal{L}$: every FADO and ARANYANI number in Section ?? is produced by minimising intersectional DP, and we report accuracy alongside the intersectional and marginal DP violations. The companion equalised-odds objective $\mathcal{L}_{EO}$ — which maintains a separate class-conditional set of running statistics (one per class per node) and adds a per-class penalty term, roughly doubling the per-step fairness cost — is the subject of an ongoing EO-objective sweep whose results are deferred (Section ??); we keep its full derivation in Section ?? because the same controller and pipeline drive it unchanged. Symbols and concrete values for λ , δ , and the remaining hyperparameters are listed in Table ?. To keep the update memory-efficient, the fairness gradient state is maintained as five running tensors—per-node weight and bias gradient accumulators, a per-class output accumulator, a subgroup count, and a per-class subgroup count—initialised once at the start of the stream and updated in place via an exponential moving average on every observed sample. The fairness contribution to the gradient is then assembled in closed form from these running statistics and added to the predictive gradient before the optimiser step. This design preserves the parameter-isolation property of the original Aranyani formulation [?]: each fairness term depends only on local node parameters, which keeps the per-step cost linear in the number of nodes and removes any need to store past samples. Predictive and fairness metrics are computed over the rolling window of size W (Table ??); using the same window for the reported metric and for the regulariser ensures that the quantity exposed to the reader

is exactly the quantity the model is being trained against, with per-step fairness evaluation in $O(W)$ rather than $O(t)$.

4.3 Drift Detection and Reaction Controller

The controller observes the online error stream $e_t = 1[\hat{y}_t \neq y_t]$ through two independent ADWIN detectors—a pre-warm detector D_w and a confirmation detector D_c —parameterised by significance levels δ_w and δ_c . The pre-warm detector is paired with a label-noise guard that excludes a sample from the change signal whenever an incorrect prediction is emitted with post-softmax confidence above ν , preventing single confident-but-wrong samples from spuriously activating the controller. A minimum of m_0 samples per detector is enforced before the first activation to avoid early-stream false positives.

When D_w fires, the optimiser learning rate is multiplied by β_w over the Δ -step horizon, pre-warming the parameters for a potential change. When D_c subsequently confirms a drift, three actions are taken atomically: (i) the learning rate is set to $\beta_c \cdot \eta_0$; (ii) the soft-routing temperature of every tree is set to τ_{drift} , sharpening routing decisions while the ensemble re-fits; and (iii) both detectors are reset and a cooldown of C samples is enforced before a new event can fire. Recovery is signalled when the rolling accuracy returns to the pre-drift baseline (estimated over the preceding window of W samples), after which τ is annealed back toward its steady-state value at rate $\Delta\tau$ per step and η decays linearly to η_0 over Δ steps. Concrete values for δ_w , δ_c , β_w , β_c , Δ , τ_{drift} , $\Delta\tau$, C , ν , and m_0 are reported in Table ?.

Table 1: Datasets.

Dataset	Size	Target	Protected attribute	Split / stream
COMPAS	7,214	Two-year recidivism	Race $\times$ Sex (4 groups)	joint 70/30 stratified train/test (5,049/2,165); two injected concept-drift scenarios on test stream
Folktables (ACS Income, CA)	187,474 187,475	+Income > \$50K	Sex $\times$ Race (4 groups)	joint Train 2015; test 2017–2018; both splits uniformly subsampled to 10% per seed

The two-stage design balances a confirmation-only reaction (too slow under sudden drift) against pre-warming without a gate (amplifies noise); combined with the label-noise guard, it separates genuine distributional change from confident-but-wrong predictions, the dominant false-positive mode we observed on the held-out tuning subset (Section ??).

4.4 Datasets and Preprocessing

Table ?? summarises the two datasets. COMPAS provides a controlled, low-dimensional testbed on which we can inject synthetic concept-drift perturbations of a known form on the test stream; FOLKTABLES (ACS Income, CA) supplies a large natural cross-year shift between 2015 (training) and 2017–2018 (prequential test). Both datasets are processed through a shared tabular pipeline: numerical features are standardised using training-set means and standard deviations, categorical variables are one-hot encoded, and (for COMPAS) the long-tail `c_charge_desc` column is frequency-encoded with rare categories collapsed below a count threshold. The encoder is fitted on the training split alone and reused unchanged for the test stream, so the streaming protocol remains free of test-time information leakage.

4.5 Concept-Drift Scenarios

We adopt complementary strategies on the two streamed datasets. On COMPAS the original distribution is approximately stationary, so we inject a synthetic, parametrically controlled concept-drift perturbation on the test stream: this isolates the abrupt label-correlated subpopulation-shift regime that the dual-ADWIN controller is designed to detect, and lets us read its effect off the prequential curves under a known ground truth. On FOLKTABLES (ACS Income, CA), training on 2015 and evaluating prequentially on 2017–2018 already provides a real, multidimensional cross-year shift driven by genuine demographic and economic change; injecting additional synthetic perturbations on top of it would conflate the controlled and natural signals and obscure the source of any observed degradation. We therefore use FOLKTABLES as an unmanipulated natural-drift testbed and reserve the controlled scenarios below for COMPAS.

Each scenario combines (i) a row-level edit of the race categorical feature in the test stream with (ii) a stochastic flip of the `two_year_recid` label on the same edited rows. The label flip is sampled independently per edited row as Bernoulli(0.5), so the flipped subset is a random half of the affected subpopulation rather than every row. The choice of $p = 0.5$ is a design decision of the testbed *a priori*, not a value tuned against the test-stream metrics: at $p = 1.0$ the flip is deterministic and the model can simply learn the inverted rule, making the drifted regime trivially separable in

the opposite direction; at $p = 0.0$ no label change occurs and the perturbation reduces to virtual covariate drift. Bernoulli(0.5) is the maximally entropic midpoint at which the post-drift labelling is unlearnable for the affected subpopulation, ensuring the scenario tests recovery rather than re-learning. Pairing the categorical edit with the per-subgroup label flip moves $P(y | x)$ as well as $P(x)$ for the affected subpopulation, yielding genuine concept drift [?] rather than purely virtual covariate drift.

The test stream is partitioned into *Warmup* (0–30%, ~ 650 samples), *Drift* (30–70%, ~ 865), and *Recovery* (70–100%, ~ 650); the edit-and-flip is applied only during the drift phase. The two active scenarios are: *no_drift*, the original distribution preserved throughout (any drift event fired here is a false positive of the detector); and *abrupt_race*, a symmetric AA $\leftrightarrow$ Caucasian swap sustained across the drift phase with a Bernoulli(0.5) `two_year_recid` flip on the same rows.

4.6 Baselines

We compare FADO against three streaming baselines that span the accuracy/fairness/adaptivity design space:

- **ARF** [?]: the Adaptive Random Forest from the river library, used as an accuracy-oriented streaming reference that handles drift through per-tree warning and drift detectors and background trees, but without any fairness mechanism;
- **RFR**: a sharpness-aware reweighting variant that applies the RFR optimiser [?] (sharpness-aware minimisation over a fairness-reweighted loss); this provides a fairness-aware static baseline without adaptive drift handling;
- **Aranyani** [?]: the bare ARANYANI forest with the node-level fairness regulariser active but the dual-ADWIN detector and reaction controller disabled. This is the strict controller-ablation of FADO — same base learner, same loss, same rolling-window stats, same seed-pinned data ordering — so any FADO-vs-ARANYANI gap is attributable to the controller alone.

All baselines are evaluated under the same batch-size-one test-then-train prequential protocol as FADO so that comparisons are not confounded by data ordering or by access to look-ahead information. The global RNG (NumPy, Python random, and TensorFlow) is pinned per seed before each model is instantiated, so the data split, the forest initialisation, and the training trajectory are bit-identical across methods that share a seed—paired statistical tests therefore compare points that differ *only* in the method under evaluation.

4.7 Hyperparameter Selection

Table ?? lists the fairness-regulariser and controller hyperparameters with their Folktables defaults and the COMPAS overrides. For the base learner we use $K = 4$ trees of depth $h = 4$ with base learning rate $\eta_0 = 2 \times 10^{-3}$, slightly larger than the ARANYANI defaults to handle Folktables’ higher dimensionality. The non-obvious overrides are justified below.

The per-dataset overrides follow from a single closed-form scaling argument: ADWIN’s cut threshold scales as $\epsilon_{\text{cut}} = \sqrt{(1/2m) \ln(4/\delta)}$, so shorter streams need a looser δ and a proportionally shorter window W , LR horizon Δ , and activation count m_0 to react within a

Table 2: FADO hyperparameters. Dash = COMPAS inherits the FOLKTABLES default.

Symbol	Folktables	COMPAS	Role
<i>Fairness regulariser</i>			
λ	1.0	–	Fairness-penalty weight
δ	0.1	–	Quadratic/linear transition point in ϕ_δ
γ	0.9	–	EMA decay of running fairness statistics
W	1000	250	Rolling window for $\mathbb{E}[\cdot]$ and reported DP
<i>Drift controller</i>			
δ_w	1×10^{-5}	1×10^{-3}	ADWIN significance (pre-warm detector)
δ_c	2×10^{-2}	2×10^{-1}	ADWIN significance (confirmation detector)
β_w	5	–	LR multiplier on warning
β_c	10	–	LR multiplier on confirmed drift
Δ	3000	600	LR decay horizon back to η_0
τ	1.0	–	Steady-state soft-routing temperature
τ_{drift}	0.1	–	Temperature set on confirmed drift
$\Delta\tau$	2×10^{-3}	–	Per-step temperature anneal back to τ
C	200	–	Cooldown (samples) between consecutive events
ν	0.7	–	Label-noise confidence guard
m_0	30	20	Min. samples per detector before first activation

single drift phase — which is why the $\sim 2k$ -sample COMPAS stream loosens δ_c by two orders of magnitude and drops W to 250 relative to FOLKTABLES. The fairness weight λ is selected per dataset on a held-out portion of the training split alone — the test stream is never consulted — via a compact NSGA-II search on FOLKTABLES and a one-dimensional grid on COMPAS; both converge to $\lambda = 1$. The remaining per-step knobs are stream-length-independent and held fixed across datasets. Appendix ?? gives the full per-knob derivation, including the label-noise guard $\nu = 0.7$ that suppresses the dominant false-positive mode.

4.8 Evaluation Protocol

All methods are evaluated under a strict prequential, batch-size-one, test-then-train protocol: at each step the model emits $\hat{y}_t$, the prequential metrics are updated, and only then is the parameter update performed using the just-observed (y_t, a_t) . We report mean and standard deviation over 30 seeds. The primary metrics are

prequential accuracy and the *intersectional* demographic-parity violation (DP) over the four joint sex $\times$ race subgroups, computed over the same W -sample rolling window that drives the fairness regulariser (Table ??). As transparency diagnostics we additionally report the two *marginal* per-attribute DP violations (DP_{sex} , DP_{race}), each obtained by projecting the joint subgroup index onto a single attribute and applying the same DP definition over the full prediction stream. For COMPAS, each (seed, model) cell produces one prequential value per scenario over the full test stream; the two scenarios are then averaged within each seed before aggregation. For FOLKTABLES, each (seed, model) cell produces one prequential value over the per-seed 10% subsample of the 2017–2018 test stream. In both cases the table reports mean $\pm$ standard deviation across the 30 seeds.

For each metric column, paired-by-seed comparisons of FADO against each baseline are performed using the two-sided Wilcoxon signed-rank test; p -values are corrected within each column using the Holm–Bonferroni step-down procedure across the three baselines. The procedure follows the prequential-benchmark recommendations of Demšar [?]; the rationale for choosing paired Wilcoxon over the Welch two-sample t -test is discussed in Appendix ??.

4.9 Reproducibility

The full implementation, configuration files, seed manifests, and per-seed result files are released at <https://github.com/TiagoSilva35/Aranyani2.0>. The repository’s README.md reproduces Table ?? and lists, for each dataset, the exact command-line invocation that produced every reported row (per-seed sweep, paired Wilcoxon + Welch significance run, summary aggregation). Running the commands as listed regenerates the same per-seed results.json files that the reported means, standard deviations, and Holm-corrected p -values are computed from.

5 Results and Discussion

We report the intersectional (sex $\times$ race) setting on the natural cross-year shift of FOLKTABLES (Table ??), as mean $\pm$ std over 30 seeds, with the demographic-parity regulariser as the optimisation target. The table gives accuracy, the intersectional DP that FADO optimises, and the two marginal per-attribute DP diagnostics. The controlled abrupt-shift COMPAS testbed under intersectional grouping, and the equalised-odds objective, are pending the companion sweep (Section ??); the COMPAS rows of Table ?? are marked accordingly.

5.1 RQ1: To What Extent Does Concept Drift Affect Group Fairness?

The FOLKTABLES cross-year shift inflates the intersectional fairness violation of both external streaming baselines (Table ??). ARF, which carries no fairness mechanism, lands at intersectional DP = 0.1497 on the 2017–2018 stream against FADO’s 0.0688. RFR, which is fairness-aware via sharpness-aware reweighting fitted on the 2015 training data, nonetheless drifts to DP = 0.1427 on the same test stream — more than double FADO’s — despite enforcing fairness at training time, which directly illustrates the static-fairness failure mode (L1 in Section ??). Both gaps are highly significant (paired Wilcoxon Holm-corrected $p < 0.001$). Crucially, the controller-free

Table 3: Prequential results under *intersectional* (sex $\times$ race) grouping, mean $\pm$ std over 30 seeds. DP is the intersectional demographic-parity violation over the four joint subgroups (the objective FADO optimises); DP_{sex} and DP_{race} are the marginal per-attribute diagnostics. Bold = best mean per metric within each dataset; superscripts on the FADO row = paired Wilcoxon Holm-corrected significance vs every baseline (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$); columns with no marker have mixed per-baseline directions and are dissected in the text. Equalised-odds results are deferred to the EO-objective study (Section ??).**

Dataset	Model	Accuracy $\uparrow$	DP $\downarrow$	DP _{sex} $\downarrow$	DP _{race} $\downarrow$
FOLKTABLES	FADO	0.7979 $\pm$ 0.0066	0.0688 $\pm$ 0.0057***	0.0234 $\pm$ 0.0064***	0.0243 $\pm$ 0.0051***
	Aranyani	0.8005 $\pm$ 0.0054	0.0727 $\pm$ 0.0050	0.0283 $\pm$ 0.0036	0.0294 $\pm$ 0.0035
	ARF	0.7446 $\pm$ 0.0107	0.1497 $\pm$ 0.0144	0.0735 $\pm$ 0.0126	0.0620 $\pm$ 0.0070
	RFR	0.7779 $\pm$ 0.0044	0.1427 $\pm$ 0.0073	0.0724 $\pm$ 0.0059	0.0591 $\pm$ 0.0048
COMPAS	FADO	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	Aranyani	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	ARF	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	RFR	[TODO:]	[TODO:]	[TODO:]	[TODO:]

ARANYANI baseline lands close to FADO (DP = 0.0727), so the bulk of the gap to ARF and RFR is contributed by the *in-processing* fairness regulariser, not by the controller.

A finding unique to the intersectional view is that the disparity drift leaves behind is *concentrated in the joint subgroups*. Measuring each method’s intersectional DP against its worst marginal DP, the unconstrained and statically-fair baselines sit at a ratio of ≈ 2.0 (ARF 0.1497/0.0735, RFR 0.1427/0.0724), whereas the in-processing fair learners are *higher* still — FADO 0.0688/0.0243 ≈ 2.8 , ARANYANI 0.0727/0.0294 ≈ 2.5 — because they scrub the marginal axes so effectively that their residual unfairness is proportionally more intersectional. A marginal-only audit would therefore have declared FADO “very fair” (DP_{sex}=0.0234, DP_{race}=0.0243) while missing a joint-subgroup gap roughly 2.8 $\times$ larger: the fairness-gerrymandering construction of [?] manifesting on real data, and the empirical justification for optimising the intersectional objective directly rather than the per-axis marginals.

5.2 RQ2: Can Adaptive Mechanisms Preserve Fairness While Maintaining Utility?

FADO attains the best mean intersectional DP and the best marginal DP on both axes against both external streaming baselines (ARF and RFR), and beats them on accuracy as well (Table ??). On FOLKTABLES this manifests as +5.3pp accuracy over ARF and +2.0pp over RFR, a $\sim 54\%/ \sim 52\%$ intersectional-DP reduction, and a corresponding reduction on both marginal axes; all twelve FADO-vs-{ARF,RFR} comparisons across the four reported metrics reach Holm-corrected paired Wilcoxon $p < 0.001$, and every one of them is also significant under the more conservative Welch t -test. The RQ2 answer against the accuracy-only and the fairness-without-adaptation baselines is therefore unambiguous: an adaptive in-processing fairness-aware learner produces the highest mean on accuracy and the lowest mean on every DP statistic, intersectional and marginal, with every win highly significant under both tests.

The FADO-vs-ARANYANI comparison isolates the contribution of the reaction controller itself: the two models share the identical base

learner, fairness loss, rolling-window statistics, and seed-pinned data ordering, so the gap between the two rows is attributable solely to the dual-ADWIN controller. The intersectional setting surfaces a controller benefit that the single-attribute view did not: the controller reduces intersectional DP by 0.38pp in FADO’s favour, and — unlike the sub-0.3pp differences typical of this gradual-shift comparison — this reduction is significant under *both* the paired Wilcoxon ($p < 0.001$) and the Welch t -test ($p < 0.01$), as is the corresponding 0.49pp/0.51pp reduction on the marginal sex and race axes (both $p < 0.001$ under both tests). The single cost is a 0.26pp accuracy deficit in ARANYANI’s favour, which is flagged by the sign-consistency Wilcoxon ($p < 0.001$) but is *not* distinguishable under Welch ($p = 0.10$) because it sits well inside the seed-to-seed spread (Appendix ??). The interpretation is that the larger joint-subgroup target gives the spike-then-anneal controller more fairness signal to act on than any single marginal axis does: optimising over four cells rather than two amplifies the per-step fairness gradient the controller modulates, so the controller’s contribution becomes measurable even on the smooth cross-year shift where the single-attribute controller was statistically tied. RQ2 is therefore answered affirmatively: FADO wins decisively against the accuracy-only and static-fair baselines, and the reaction controller adds a both-tests-significant intersectional-DP improvement over the controller-free base learner at a Welch-negligible accuracy cost.

6 Conclusion and Future Work

We introduced FADO, which wraps the ARANYANI fair oblique decision forest (the inherited in-processing base learner) with an ADWIN-driven dual-threshold detector and a reaction controller that spikes the learning rate and sharpens the soft-routing temperature on confirmed drift and anneals both back during recovery; the contribution is the drift-detection-and-response machinery and its intersectional fairness instantiation.

Empirically, on FOLKTABLES under intersectional (sex $\times$ race) grouping over 30 seeds, optimising demographic parity, FADO more than halves the intersectional DP of ARF and RFR (all twelve comparisons at Holm-corrected $p < 0.001$ under both tests) and, unlike the single-attribute setting, beats the controller-free ARANYANI base learner by a both-tests-significant margin (Wilcoxon $p < 0.001$, Welch $p < 0.01$) at a Welch-negligible 0.26pp accuracy cost — the larger four-cell objective gives the controller more fairness signal to act on (RQ2). Residual disparity, for *every* method including the fair learners, stays concentrated in the joint subgroups (intersectional-to-marginal ratio $\approx 2\text{--}3$), reproducing the fairness-gerrymandering construction [?] on real data and justifying intersectional over marginal optimisation (RQ1).

The most immediate extension is the *equalised-odds objective*: the EO regulariser (Appendix ??) is driven by the same controller, and a companion sweep optimising EO directly — with the intersectional COMPAS testbed left as placeholders in Table ?? — is underway. Further directions include time-resolved fairness metrics, extension to $K > 2$ groups, and a component ablation of the controller’s branches.

A Equalised Odds: Metric and Regulariser

Equalised odds (EO) is the label-conditional fairness notion that compares group prediction rates separately within each true-label class. For each $y \in \{0, 1\}$, $\hat{p}_{a|y} = \mathbb{E}[\hat{y} \mid a, y]$ is the positive-prediction rate of group a within class y and $\bar{p}_y = \frac{1}{|\mathcal{A}|} \sum_{a'} \hat{p}_{a'|y}$ its cross-group mean; the EO violation is the worst-case deviation across both classes,

$$\text{EO} = \max_{y \in \{0,1\}} \max_{a \in \mathcal{A}_t} |\hat{p}_{a|y} - \bar{p}_y|,$$

and restricting the outer maximisation to $y=1$ yields equal opportunity. EO disentangles bias from class imbalance and is informative when base rates differ across groups.

The companion equalised-odds objective referenced in Section ?? conditions each running statistic on the true label. For $c \in \{0, 1\}$:

$$\bar{n}_{ij,a|c} = \mathbb{E}[n_{ij}(x) \mid y = c, a], \quad \bar{n}_{ij|c} = \frac{1}{|\mathcal{A}|} \sum_{a' \in \mathcal{A}} \bar{n}_{ij,a'|c},$$

where $\bar{n}_{ij,a|c}$ is the group-and-label-conditional average routing decision at node (i, j) (the per-node analogue of $\hat{p}_{a|y}$) and $\bar{n}_{ij|c}$ is its cross-group mean within class c . The corresponding node deviation and EO regulariser are:

$$F_{ij,a}^{(c)} = \bar{n}_{ij|c} - \bar{n}_{ij,a|c},$$

$$\mathcal{L}_{\text{EO}} = \lambda \sum_{(i,j)} \frac{1}{2|\mathcal{A}|} \sum_{c \in \{0,1\}} \sum_{a \in \mathcal{A}} \phi_{\delta} \left(F_{ij,a}^{(c)} \right),$$

which differs from the DP regulariser only by the additional sum over c and the corresponding $\frac{1}{2}$ normalisation; the rest of the structure (per-node penalty, group average, ϕ_{δ} surrogate) is identical. The same dual-ADWIN controller and prequential loop (Algorithm ??) drive it without modification.

B Per-Dataset Hyperparameter Calibration

This appendix expands the per-dataset overrides summarised in Table ?? and referenced in Section ?. All per-dataset overrides follow from one closed-form scaling argument and one pilot-calibration observation. ADWIN's cut threshold scales as $\varepsilon_{\text{cut}} = \sqrt{(1/2m) \ln(4/\delta)}$, so for fixed δ the minimum detectable accuracy deficit grows as $1/\sqrt{m}$: shorter streams need a looser δ to detect drifts of equivalent magnitude. On FOLKTABLES the long stream tolerates a tight bound ($\delta_w=10^{-5}$, $\delta_c=2 \times 10^{-2}$, sub-1 pp floor); on COMPAS the same δ_c yields an ~ 9 pp floor that exceeds the actual deficits during the ~ 865 -sample drift phase, so we loosen by two orders of magnitude ($\delta_w=10^{-3}$, $\delta_c=2 \times 10^{-1}$). The LR decay horizon Δ and the minimum activation count m_0 are scaled proportionally to stream length so the controller can complete a full spike/decay cycle and so it does not fire on early-stream noise. The rolling window W is set so it turns over within a single drift phase: $W = 1000$ covers 0.27% of the FOLKTABLES stream but would cover 46% of the COMPAS stream, so we drop W to 250 on COMPAS, matched to ADWIN's ~ 200 -sample bucket scale. The fairness weight λ is the only knob with a non-geometric calibration and is selected per dataset on a held-out portion of the training split alone — the test stream is never consulted. On FOLKTABLES, a compact NSGA-II procedure (population 8, 4 generations) jointly minimises prequential 1-Accuracy and DP on the held-out training subset and picks from the first Pareto front the candidate that minimises the normalised sum of objectives;

this yields $\lambda = 1$. On COMPAS the held-out training subset is too small to support a multi-generation NSGA-II reliably, so we run the same prequential-on-training procedure over a one-dimensional grid $\lambda \in \{0.1, 1, 10\}$ and select the value that minimises the same combined objective on the held-out training subset; this also yields $\lambda = 1$. The two datasets therefore converge on the same value: at $\lambda = 10$ the per-step fairness gradient dominates the classification gradient and the in-sample DP/accuracy product on the held-out subset degrades; at $\lambda = 0.1$ the regulariser contribution becomes too small for the in-sample DP estimate to track group statistics responsively. The remaining knobs are stream-length-independent per-step quantities and are held fixed across both datasets at the Table ?? values. The surrogate transition $\delta = 0.1$ is inherited from ARANYANI and gives a quadratic regime for routing deviations below 10pp and a linear regime above, so single-sample gradient noise is treated smoothly while outlier deviations are kept bounded. The EMA decay $\gamma = 0.9$ corresponds to an effective half-life of ~ 7 samples, fast enough to track per-phase distributional change but slow enough to suppress single-sample noise. The asymmetric LR multipliers $\beta_w=5$ and $\beta_c=10$ enforce a 2:1 confirmation/warning ratio: warnings produce a measurable pre-warm without committing the model to a regime change, and only confirmation triggers the full spike. The drift-time temperature $\tau_{\text{drift}} = 0.1$ (a tenfold reduction from steady-state $\tau = 1.0$) makes routing essentially hard for the duration of the spike, so the ensemble can commit to a different leaf assignment within a few dozen samples rather than waiting for soft-routing weights to average over the inverted regime; the per-step anneal $\Delta\tau = 2 \times 10^{-3}$ returns τ from 0.1 to 1.0 over 450 steps, which fits inside the $\Delta = 600$ LR decay horizon on COMPAS and synchronises temperature recovery with the LR schedule. The cooldown $C = 200$ prevents re-firing during the controller's own response and is matched to ADWIN's ~ 200 -sample bucket scale. The label-noise confidence guard $\nu = 0.7$ excludes high-confidence-but-wrong predictions from the pre-warm error stream: on the same held-out training subset used for λ calibration, single samples where the forest emitted a high-confidence prediction that disagreed with the noisy ground truth were the dominant false-positive mode, and a 0.7 threshold removed them without measurably affecting in-sample true-positive detection of distributional change.

C Choice of Statistical Test

We report the paired Wilcoxon signed-rank test as the primary statistic, following the prequential-benchmark recommendations of Demšar [?]: it makes no parametric assumption about the per-seed metric distribution, and it exploits the within-seed pairing produced by the shared data ordering (Section ?), which is precisely the structure that makes the Welch two-sample t -test discard relevant information. In the intersectional setting the two tests agree on every DP comparison in Table ?? — including the FADO-vs-ARANYANI controller-isolation gap, which clears Welch at $p < 0.01$ (intersectional) and $p < 0.001$ (both marginals) — so the fairness conclusions do not hinge on the choice of test. The two tests disagree on exactly one cell: the 0.26pp FADO-vs-ARANYANI accuracy deficit, which is small in magnitude but consistent in sign. There Welch is non-significant ($p = 0.10$, the gap is within the standard deviation of either model in isolation) while paired

Wilcoxon flags it because the sign of the per-seed difference is consistent across 25+ of the paired seeds. Both numbers are correct; they answer different questions. Welch asks whether the two models' marginal distributions differ; paired Wilcoxon asks whether

the controller systematically shifts the per-seed outcome in one direction. We report both, and note that the accuracy cost of the controller is negligible under the distributional reading and merely systematic-but-tiny under the paired reading.

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.